

[SNU 2021 Spring #8, Robbins-Siegmund Lemma.

Let $\{V_k\}, \{S_k\}, \{U_k\}$ be non-negative real sequences satisfying

$$V_{k+1} \leq V_k - S_k + U_k \text{ for all } k \in \mathbb{N} \text{ and } \sum_{k=0}^{\infty} U_k < \infty$$

Then, $S_k \rightarrow 0$ and $\{V_k\}$ has a limit as $k \rightarrow \infty$.

Proof. observe that

$$V_{k+1} - V_k + S_k \leq U_k \text{ for all } k \in \mathbb{N}.$$

Therefore, given $N > 1$,

$$\sum_{k=1}^N (V_{k+1} - V_k + S_k) = V_{N+1} - V_1 + \sum_{k=1}^N S_k \leq \sum_{k=1}^N U_k \leq \sum_{k=1}^{\infty} U_k$$

thus
$$\sum_{k=1}^N S_k \leq \sum_{k=1}^{\infty} U_k + V_1 - V_{N+1} \leq \sum_{k=1}^{\infty} U_k + V_1 < \infty,$$

So $\sum_{k=1}^{\infty} S_k$ converges, $S_k \rightarrow 0$ as $k \rightarrow \infty$.]

Meanwhile, define

$$W_k = V_k - \sum_{i=1}^{k-1} U_i \quad (k \geq 2).$$

Then, for any $k \geq 2$,

$$W_{k+1} - W_k = V_{k+1} - V_k - U_k \leq -S_k \leq 0, \text{ so } \{W_k\} \text{ is monotone decreasing.}$$

Moreover, given $k \geq 2$,

$$W_k = V_k - \sum_{i=1}^{k-1} U_i \geq -\sum_{i=1}^{k-1} U_i \geq -\sum_{i=1}^{\infty} U_i$$

thus, W_k is bounded below, so converges. Now,

$$V_k = \sum_{i=1}^{k-1} U_i + W_k$$

Converges, being sum of convergent sequences.